NetWalkRank: Cancer driver gene prioritization in multiplex gene regulatory networks by a random walk approach

Fatemeh Keikha, Lingyu Li, Wai-Ki Ching, Zhi-Ping Liu

***Abstract*—Finding and prioritizing cancer driver genes (CDGs) that disrupt normal cell functionality and contribute to cancer occurrence and development is a significant challenge in oncology. Integrating multiple information pertaining to the characteristics of each gene at different stages of the disease and incorporating multiple steps as individual layers in the model provides a more comprehensive understanding of each node or gene. Thus, it is reasonable to organize them into multiplex gene regulatory networks (GRNs). In this work, we present a network-based framework called NetWalkRank, for prioritizing CDGs in the multiplex GRNs with gene expression profiling data. The framework applies the concept of network propagation to calculate the relative impact of each gene in spreading abnormality throughout the multiplex GRNs. It was employed to give priority to the driver genes of hepatocellular carcinoma (HCC) in humans. The performance of NetWalkRank was demonstrated through the ranks and classifications assigned to the known CDGs, which validated its effectiveness. To showcase the predictive capabilities of our proposed framework, we trained a random forest model that utilizes the obtained scores to accurately predict CDGs. We compared the advantage and efficiency of our method with other well-known driver gene ranking methods through numerical experiments. The findings show that the usage of GRNs across various steps of multiplex networks in prioritizing and predicting CDGs is significant, as demonstrated by the efficiency and effectiveness of NetWalkRank. All data and source code used in this study are available at https://github.com/zpliulab/NetWalkRank.**

***Index Terms*—Cancer driver gene prioritization, Gene regulatory networks, Multiplex networks, Differential mutual information, Random walk model.**

I. Introduction

CANCER, characterized by cellular abnormalities and uncontrolled growth due to gene mutations, is a major cause of global mortality and poses significant health risks [1]–[4]. Among its types, hepatocellular carcinoma (HCC) is particularly challenging for early diagnosis and effective treatment [5]. Given its high fatality, identifying genes that drive HCC development and progression is crucial for improving patient outcomes [6]. Gene regulatory networks (GRNs) systematically model the interactions among multiple genes that collaborate to promote cancer [7], [8]. However, mutations can disrupt normal cellular functions, leading to cancer [9]. While driver mutations trigger tumor growth by conferring selective advantages, non-driver mutations do not impact progression [10], [11]. Therefore, identifying cancer driver genes (CDGs) is vital, as they are essential for effective diagnosis and treatment [12].

Conventional methods analyze genetic aberration frequencies across cancer cohorts [13]–[15], identifying major driver genes with higher mutation rates [16]. Moreover, tools like MutsigCV [15], MDPFinder [17], and iPAC [18] integrate gene expression and mutational data to identify CDGs. Methods such as OncodriveCLUST [19] focus on mutation clustering, while functional annotation approaches like OncodriveFM [20] assess the impact of mutations on proteins. In contrast, recent research methods have concentrated on identifying key mutated genes related to cancer [21]–[23], using computational methods to distinguish between cancer driver mutations and passenger mutations. The prognostic impact of these mutations is influenced by the presence of other driver mutations [10], making the identification and prioritization of CDGs from the vast human genome essential for understanding cancer pathogenesis [24]. Gene prioritization condenses thousands of genes into a manageable list [25], with most methods relying on computational approaches [26]. These approaches expedite the identification of therapeutic targets, support precision medicine, and enhance patient outcomes by uncovering driver mutations through statistical and machine learning techniques. All of these conventional and recent methods highlight the need for more comprehensive approaches in cancer research, paving the way for more advanced strategies.

Cells are composed of diverse molecules that form intricate and dynamic interaction networks [27], [28]. Genetic aberrations can alter these networks by impacting or eliminating nodes or connections, or by modifying biochemical characteristics [29]–[32]. The vast amount of cancer genomics data offers significant opportunities to understand tumor initiation and progression at a network or systems level [33]. Cancer is recognized as a multifaceted disease with alterations at the network and pathway levels rather than solely due to individual point mutations [34]. Traditional methods based on frequency analysis and functional annotation are inadequate for identifying rarely mutated driver genes [16]. Consequently, there has been considerable enthusiasm for network-based and pathway-based approaches to prioritize driver mutations, as these mutations often impact cellular signaling and regulatory pathways [34], [35].

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In recent years, several promising methods have emerged that consider somatic mutations within the context of pathways. Network-based approaches, such as protein-protein interaction (PPI) networks, identify genes with strong interconnections and functional relationships with known driver genes, providing valuable insights into cancer's molecular mechanisms. For instance, DawnRank [36] uses mutational and transcriptome data with molecular interaction networks and employs the PageRank algorithm to prioritize driver genes within an individual patient. Similarly, DriverNet [37] examines genomic patterns and transcriptional data to identify driver mutations, while iMaxDriver [38] focuses on gene networks to identify CDGs in the Transcription Factors category using the influence maximization approach. However, existing methods often overlook the significance of neighboring nodes or fail to consider topological characteristics. Thus, there is a need for novel approaches that integrate graph topological characteristics and propagation tendencies to identify CDGs effectively. Incorporating multistep gene expression data as layers in the analysis facilitates a comprehensive exploration of dynamic changes during cancer progression, leading to an enhanced understanding of molecular pathways involved in tumor development and improving the prioritization of CDGs.

In the present study, we developed a novel framework called NetWalkRank that prioritizes driver genes in GRNs using gene expression profiling data through a multiplex network approach and an improved random walk algorithm. NetWalkRank combines the power of the PageRank algorithm and random walks to assess the importance of driver genes based on their connectivity within and across multiplex network structures, thereby constructing the cancer regulatory network at different stages of disease development. We demonstrated the effectiveness of NetWalkRank in prioritizing CDGs in HCC by constructing a multiplex GRN and mapping gene expression data onto it. We then employed a rank aggregation method to prioritize these genes based on their final values across layers while taking into account the importance of each layer.

While interactome-based approaches are widely used for disease gene identification, they are inherently influenced by knowledge bias, where well-characterized driver genes are often prioritized due to their extensive annotation in curated databases. Recent studies [39], [40] have shown that different interactomes vary in network properties and predictive performance, highlighting the potential benefits of integrating multiple sources for improved robustness. To address these challenges, NetWalkRank constructs a multiplex GRN that incorporates dynamic gene expression data across disease stages, ensuring that gene ranking reflects stage-specific molecular changes rather than relying solely on prior interaction knowledge. To highlight the advantages of NetWalkRank, we conducted a comparison of its identification results with other methods, including DawnRank [36], DriverNet [37], iMaxDriver [38], and MDPFinder [17], for identifying CDGs. We assessed the network independence of the results using three distinct PPI networks: STRING [41], InBioMap [42], and HumanNet [43]. Additionally, we tested the robustness of our method by introducing randomly added edges to the RegNetwork and evaluating NetWalkRank's performance in identifying driver genes under these conditions. Finally, we confirmed the independence of the results from gene expression data by prioritizing and categorizing driver genes using an alternative independent HCC gene expression dataset.

II. Materials and methods

A. Databases and datasets

To investigate the feasibility of using NetWalkRank to prioritize CDGs associated with HCC, a proof-of-concept study was conducted using a multi-step approach. Initially, gene expression data specific to HCC were obtained from the NCBI GEO database [44] under the entry ID GSE89377 [45] shown in Supplementary Table S1.1). The dataset consists of 90 tumor samples and 13 control samples, representing nine different stages of the disease. To organize the multi-step data, a framework of a multiplex network was implemented. The GRN for each stage of the disease was constructed using the RegNetwork database [46]. The RegNetwork database contains a comprehensive collection of gene regulatory interactions derived from multiple sources and databases, encompassing a total of 372,774 regulatory relationships involving 23,079 genes.

The main aim of this study is to assess the effectiveness of the proposed approach in prioritizing CDGs, with a specific focus on those associated with HCC. To evaluate the methodology, we compiled driver genes from four widely recognized HCC driver gene databases: Malacards [47], OncoVar [48], NCG [49], and DriverDBv3 [50]. A total of 20 driver genes were identified, consisting of genes that appeared in more than 50% of the compiled lists (shown in Supplementary Table S1.2). To assess the robustness of our method, we designed an extensive validation procedure. In each of the 1,000 experimental runs, we selected a random set of normal genes (negative samples) for comparison. The set of 20 well-known driver genes was used as positive samples for training, while the remaining known driver genes, along with a random set of negative samples in the same size, were used as the test set. This approach allows us to systematically evaluate the model's ability to distinguish true driver genes from other genes in the dataset, ensuring a comprehensive and reliable performance assessment.

Specifically, the gene expression data was integrated with the GRN by applying the differential mutual information (DMI) measure, which served as the edge weight. Supplementary Table S2.1 provides detailed information about the gene expression dataset and the GRN employed in this study. Nodes representing the same genes in different layers were connected as inter-layer interactions. As a result, a multiplex network was constructed, consisting of nine layers, 77,805 nodes and 491,386 interactions. After removing genes that lacked differentially expressed information across different stages, the resulting network contained 38,389 nodes and 88,931 edges.

B. Framework

Fig. 1 provides an overview of the NetWalkRank framework, which comprises three main parts. In Fig. 1(a), the gene

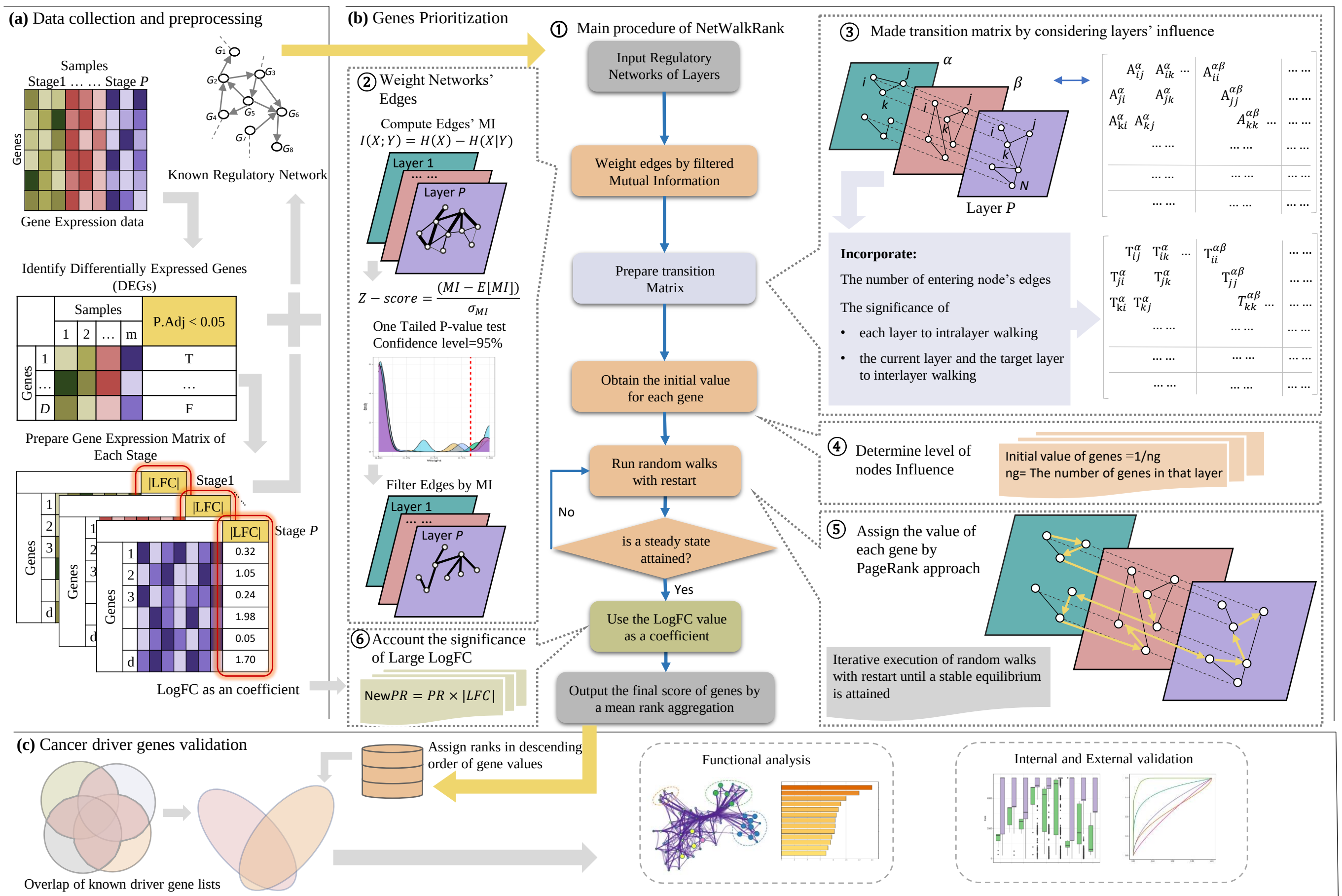

Fig. 1: The framework of NetWalkRank. (a) Data collection and preprocessing. Gene expression data and gene regulatory network were used to make a multiplex network. (b) Gene prioritization. The main procedure of NetWalkRank to prioritize driver genes contains weight and filter networks' edge, made transition matrix, run personalized PageRank, and obtain the final score for each gene. (c) Cancer driver genes validation. Some validation tests on assigned rank genes prove the reliability of the results: classification validation, and prioritization of cancer driver genes.

expression data associated with HCC, containing information from multiple stages, is obtained from the GEO database. Additionally, a GRN is constructed based on the gene expression data, forming a network representation. The multiplex network is then created by integrating the GRNs from different stages of HCC, establishing interactions across the various layers.

The core procedure of NetWalkRank begins by assigning weights to the interactions using the DMI measure, as depicted in Fig. 1(b-2). Subsequently, an improved PageRank algorithm is applied to the multiple weighted networks. To execute the algorithm on the multiplex network, a transition matrix is prepared, and the initial values of each node are determined based on the layer to which they belong (as shown in Fig. 1(b-3, b-4)). After performing random walking with restart until a steady state is reached, the final scores of the nodes are computed by incorporating the Log-Fold Change (LFC) of each node in the corresponding stage as a coefficient and employing a weighted mean aggregation method (in Fig. 1(b-5, b-6)). The performance of classification and the impact of the NetWalkRank method on the rankings of driver genes are evaluated as part of the validation study (in Fig. 1(c)).

C. Multiplex gene regulatory networks

We retrieved the gene expression data from the GEO database and applied a linear model in the Limma package [51] to analyze the data. By employing a Bayesian model, we calculated the contrasts between the disease (d) and control (c) groups to identify genes exhibiting differential expression. To prioritize the most significant differentially expressed genes, we set a threshold using the false discovery rate based on adjusted p-value (≤ 0.05). Genes that did not display differential expression across different disease stages were excluded from further analysis.

To organize the data corresponding to different disease stages, we utilized a multiplex network model. Specifically, we constructed a GRN for each disease stage by leveraging the RegNetwork database and gene expression data, which is a curated collection of regulatory interactions based on existing knowledge. For inter-layer interactions, we defined connections based on the presence of identical genes across different layers.

The edges in each layer of the multiplex network are assigned weights based on the DMI of gene expression between the control and tumor samples in that particular disease stage:

$$DMI = |MI(X_c, Y_c) - MI(X_d, Y_d)|. \quad (1)$$

The DMI measure shown in Equation (1) represents the absolute difference between the mutual information (MI) values of the control and disease states. The MI is calculated using a straightforward approach, where the variables, such as X and Y, are partitioned into bins of finite size N. A two-dimensional histogram is constructed on a scatter plot of X and Y, where each bin represents a discrete region in the joint distribution of X and Y. The number of data points falling into each bin is counted to approximate probability distributions. The MI is then estimated as follows:

$$I(X,Y) = \sum_{(i,j)} \frac{\log(p(i,j))}{p_x(i)p_y(j)}, \quad (2)$$

where $p_x(i) \approx \frac{n_x(i)}{N}$, $p_y(j) \approx \frac{n_y(j)}{N}$, and $p(i,j) \approx \frac{n(i,j)}{N}$. Here, $n_x(i)$ represents the number of points falling into the i-th bin of X, $n_y(j)$ represents the number of points falling into the j-th bin of Y, and $n(i,j)$ represents the number of points in their intersections [52].

In data pre-processing, the DMI values are normalized using the min-max method, and genes that do not exhibit differential information across different steps are removed. In each layer, the interactions with significant DMI values are retained by analyzing the frequency distribution of these MI values for each step. The z-score is utilized to evaluate the statistical significance of these gene interactions. By applying a one-sided statistical test with a confidence level of 95%, a cutoff is determined that remains consistent across all steps. This leads to dynamic changes in the edges between the steps of HCC development (refer to Supplementary Fig. S2.1).

We define the super-adjacency matrix for our multiplex network as G, which is used to represent the relationships between genes in each layer and across layers:

$$G = \begin{bmatrix} G^1 & G^{12} & \cdots & G^{1p} \\ G^{21} & G^2 & \cdots & G^{2p} \\ \vdots & \vdots & \vdots & \vdots \\ G^{p1} & \cdots & \cdots & G^p \end{bmatrix}, \quad (3)$$

where the number of layers is in $1, 2, \cdots, p$, and G^α $(1 \leq \alpha \leq p)$ in Eq. (3) is defined as follows:

$$G^\alpha = \begin{bmatrix} 0 & G^\alpha_{12} & \cdots \\ G^\alpha_{21} & 0 & \cdots \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad (4)$$

where G^α_{ij} $(1 \leq i, j \leq N)$ represents the relationships between genes within layer α, with N being the number of nodes in the gene network. If gene i regulates gene j in layer α, $G^\alpha_{ij} = 1$ otherwise is set to 0. Since GRNs are directed, each step of the GRN is represented by a directed network, even though MI itself is not directional.

Furthermore, $G^{\alpha\beta}$ $(1 \leq \alpha, \beta \leq p)$ in Eq. (3) is defined as follows:

$$G^{\alpha\beta} = \begin{bmatrix} G^{\alpha\beta}_{11} & 0 & \cdots \\ 0 & G^{\alpha\beta}_{22} & \cdots \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad (5)$$

where $G^{\alpha\beta}$ is a diagonal matrix and $G^{\alpha\beta}_{ii}$ is replacing 1 if gene i is located in both layers α and β, and otherwise it remains zero.

We obtain the weighted super adjacent matrix of our multiplex network, based on mentioned DMI and inter-layer connections. Let A be the weighted adjacent matrix:

$$A = \begin{bmatrix} A^1 & A^{12} & \cdots & A^{1p} \\ A^{21} & A^2 & \cdots & A^{2p} \\ \vdots & \vdots & \vdots & \vdots \\ A^{p1} & \cdots & \cdots & A^p \end{bmatrix} \quad (6)$$

where A^α $(1 \leq \alpha \leq p)$ in Eq. (6) is defined as follows:

$$A^\alpha = \begin{bmatrix} 0 & A^\alpha_{12} & \cdots \\ A^\alpha_{21} & 0 & \cdots \\ \vdots & \vdots & \vdots \end{bmatrix}, \quad (7)$$

with A^α_{ij} presenting the DMI value between gene i and gene j in layer α, and $A^{\alpha\beta}$ is equal to $G^{\alpha\beta}$. Fig. 1(b-3) provides a visual representation of the conversion process of the multiplex network into a matrix.

D. *Improved personalized PR algorithm on the weighted multiplex network*

The influence of nodes is assessed using an enhanced personalized PR algorithm that incorporates constraints on information flow within and between layers of the network. By utilizing the DMI values, we constructed weighted matrices for our multiplex network (referred to as matrix A). Additionally, we created the super adjacency matrix (designated as matrix G) by replacing values greater than zero with 1, which provides a visual representation of the multiplex network structure.

The prioritization of driver genes from the homogeneous multiplex network using the improved PR algorithm (see Algorithm 1). Algorithm 1 provides a detailed overview of the algorithm introduced in this study. It describes the behavior of random particles positioned at each node within the network, which have the option to either remain within their current layer or transition to other layers based on the transition matrix of the homogeneous multiplex network.

Following the construction of the weighted matrix for a multiplex network, the next step involves constructing the weighted transition matrix for a multiplex network using the weighted matrix and the super adjacency matrix. Let T be the transition matrix in the homogeneous multiplex network:

$$T = \begin{bmatrix} T^{11} & T^{12} & \cdots & T^{1p} \\ T^{21} & T^{22} & \cdots & T^{2p} \\ \vdots & \vdots & \vdots & \vdots \\ T^{p1} & \cdots & \cdots & T^{pp} \end{bmatrix} \quad (8)$$

where $T^{\alpha\alpha}$ refers to the intro-subnetwork transition matrix in layer α and $T^{\alpha\beta}$ is the inter-subnetwork transition matrix between layer α and layer β. Not all nodes exist in all layers due to the remaining nodes that are on sides of a connection with MI higher than the threshold. So when the random walker is in layer α, he can jump to layer β or stay in layer α.

If he is on the node connecting to another layer, he can jump to them or move to the other node in the current layer. We define jumping probabilities for the random walker. The proposed method incorporates the significance of each layer in determining the probability of transferring genetic information between the genes within that layer of network. In addition, the likelihood of a random walker transitioning to another layer is influenced by both the importance of the current layer and the target layer, which are quantified using coefficients based on the weighted values of the edges within each layer. Let L^α be the impact coefficient of layer α:

$$L^\alpha = W^\alpha / \sum_{k=1}^{p} W^k, \tag{9}$$

where k refers to a single-layer network used in the multiplex p layers, and

$$W^\alpha = \begin{cases} S_A^\alpha / S_G^\alpha, & \text{if } S_G^\alpha \neq 0, \\ 0, & \text{otherwise}, \end{cases} \tag{10}$$

where S_A^α represents the total sum of A_{ij}^α, and S_G^α represents the total sum of G_{ij}^α.

Algorithm 1 The improved personalized PR algorithm

Input: Gene expression data, GRN.
Calculation: Find differentially expressed genes, Calculate DMI for each edge in the multiplex network.
Filtering: Select edge with DMI larger than a given threshold.
Construction: Construct transition matrix T, Construct super adjacency matrix of multiplex network M.
Initialization:
Initialize Parameters:
$d = 0.3$,
$PR_0 = [PR_0^1, PR_0^2, \cdots, PR_0^p]$,
$PR_t = PR_0$,
del_p = inf,
iter = 1.
PageRank:
while del_p $> 1\mathrm{e}{-10}$ or iter < 200 **do**
$PR_t = PR$,
$PR = (1-d) \times MT^\top \times PR_t + d \times PR_0$,
del_p = $\|PR - PR_t\|_1$,
iter = iter + 1.
end while
Output: PR value of genes according to PR.

Therefore, Algorithm 1 also incorporates coefficients based on the relative importance of each layer to construct the transition matrix. Each element of T can be defined as follows. For a transition matrix in the intra-layer, such as $T^{\alpha\alpha}$ in each layer, the probability of a random walker at the i-th row and j-th column is defined as follows:

$$T_{ij}^{\alpha\alpha} = \begin{cases} L^\alpha A_{ji}^\alpha P_{ij}^\alpha, & \text{if } \sum_j G_{ij}^\alpha \neq 0, \\ 0, & \text{otherwise}, \end{cases} \tag{11}$$

where $P_{ij}^\alpha = G_{ij}^\alpha / \sum_j G_{ij}^\alpha$. The term P_{ij}^α represents the normalized weight, defined as the ratio of G_{ij}^α to the total sum of G_{ij}^α over all j, ensuring that the values are scaled relative to the overall distribution.

For the inter-layer transition matrix $T^{\alpha\beta}$, the probability of a random walker at the i-th row and j-th column is defined as follows:

$$T_{ii}^{\alpha\beta} = \begin{cases} L^{\alpha\beta} A_{ii}^{\alpha\beta} P_{ii}^{\alpha\beta}, & \text{if } \sum_j G_{jj}^{\alpha\beta} \neq 0, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

where $L^{\alpha\beta} = (L^\alpha + L^\beta)/2$, and $P_{ii}^{\alpha\beta} = G_{ii}^{\alpha\beta} / \sum_j G_{jj}^{\alpha\beta}$. The term $P_{ii}^{\alpha\beta}$ represents the normalized weight, defined as the ratio of $G_{ii}^{\alpha\beta}$ to the total sum of $G_{jj}^{\alpha\beta}$ over all j, ensuring that the values are scaled relative to the overall distribution.

Next, we present the formulation of the random walk with the restart algorithm applied to the multiplex network. In this study, we employed a constrained personalized PR approach. To incorporate prior knowledge about the genetic information flows, we defined a constraint matrix M based on the characteristics of the transition matrix T. Specifically, we construct M from the adjacency matrix of the T, where each entry represents a constrained transition that reflects the directionality and specific requirements of genetic information flow. To ensure meaningful transitions, self-loops are removed, and the matrix is column-wise normalized so that each column sums to 1 where transitions are allowed, preserving its stochastic properties.

Let PR_0 be the initial probability vector and PR_t be the vector in which the i-th element holds the probability of finding the random walker at node i at step t. The probability vector at step $t+1$ can be given by:

$$PR_{t+1} = (1-d) \times MT^\top \times PR_t + d \times PR_0, \tag{13}$$

where d is the restart probability, which represents the chance of the random walker going back to the seed nodes. The Random Walk with Restart (RWR) algorithm utilizes seeds as initial sources of information. At each restart, the algorithm restarts from the seed genes and propagates through the network based on the probability value specified by the parameter d. The initial probability vector PR_0 assigns initial weights to nodes in each layer based on the number of genes within that layer, serving as the initial starting points for the random walk, given by:

$$PR_0 = [PR_0^1, PR_0^2, \cdots, PR_0^p], \tag{14}$$

where $PR_0^\alpha = 1/n_\alpha$ and n_α represents the number of genes within layer α. In this study, we set d to 0.3 based on empirical tests using a grid search technique (see Supplementary Tables S2.2 and S2.3). After iterating for several steps, the probability will reach a steady state, which can be obtained by performing iterations until the difference between PR_{t+1} and PR_t (measured by L_1 norm of one vector) falls below a threshold, such as $1\mathrm{e}{-10}$.

After applying PR to the multiplex network, each node achieves a PR value at each level. To account for the significance of genes with large LFC values in their expression levels, if a gene's LFC value in a particular step is greater than one, the proposed method uses this LFC value as a coefficient when computing the value for that gene in the subsequent layer,

$$NewPR_i^\alpha = \begin{cases} PR_i^\alpha \times |\mathrm{LFC}_i^\alpha|, & \text{if } |\mathrm{LFC}_i^\alpha| > 1, \\ PR_i^\alpha, & \text{otherwise.} \end{cases} \tag{15}$$

In Eq. (15), PR_i^α represents the resulting value of the PR algorithm for node i in layer α, and LFC_i^α represents the log fold change of expression of that node in that step.

E. Rank aggregation

A weighted mean value aggregation method that considers the impact of each layer is utilized to derive the final score of each gene. Let $R(i)$ represent the final score of the i-th node, which is defined by

$$R(i) = \sum_{k=1}^{P} L^k NewPR_i^k, \tag{16}$$

where k refers to a single-layer network used in the multiplex p layers defined in Eq. (16), L^k represents the impact of the k-th layer of the network, as shown in Eq. (9).

The final values obtained from the algorithm are arranged in descending order, and ranks are assigned to nodes based on their positions in this sorted list. Nodes with identical values are given the same rank, which corresponds to the lowest position in the list among nodes with the same score.

III. Results and discussion

A. Prioritization of cancer driver genes

Initially, we employed NetWalkRank to determine the rankings of HCC driver genes. In Fig. 2(a), we present the boxplots that illustrate the ranks obtained through various strategies, which involve combining networks and considering specific data characteristics. Our first approach involved ranking the HCC driver genes within the regulatory network, treated as a single-layer network. This is represented by the box plot labeled 'REG', which solely takes into account the network topology. The remaining boxplots depict the results of experiments conducted on gene ranking by integrating different disease stages as a multiplex network. The boxplot labeled 'REGm' corresponds to the ranking results of HCC driver genes within the original GRN, where the MI weighting method incorporates both the network topology and gene expression profiles for gene prioritization.

In addition, we conducted NetWalkRank analysis on the multistep gene expression data of HCC mapped onto a multiplex network. Within this framework, we employed different strategies to enhance the gene prioritization process. 'REGml' represents the ranking results obtained in the multiplex network by incorporating MI to weight and filter gene connections. 'REGmlw' signifies the ranks derived from utilizing the normalized weight of each layer as a coefficient in constructing the transition matrix. Furthermore, we incorporated layer-specific information by setting the initial value of each node in PR based on the number of genes present in its corresponding layer. The ranks obtained through the improved PR algorithm with a transition matrix weighted by MI and utilizing normalized layer weights are denoted as 'REGmlwp'. Our findings indicate that when more properties of the multiple networks were considered in the prioritization process, the CDGs tended to achieve higher ranks.

The elevated ranks observed in NetWalkRank highlight its effectiveness in prioritizing cancer genes. Fig. 2(a) illustrates a gradual increase in the ranks of driver genes accompanied by decreasing mean values and standard deviations. This trend becomes more prominent as we incorporate additional properties of gene expression data and properties of the multiplex network in the experiments. Notably, the mean rank of driver genes has improved from 184 in 'REGmlwp' to 149 in 'REGmlwpFC', where the final value for each gene in a given layer is computed by utilizing the LFC value of the gene as a coefficient.

Fig. 2(b) demonstrates the impact of incorporating additional steps of disease progression in the prioritization of CDGs. By integrating the last three, six, or nine steps of HCC in NetWalkRank as layers of the multiplex network, the mean rank of driver genes has significantly improved, decreasing from 665 to 149. This enhancement highlights the importance of including more layers in the multiplex GRN, as it provides multiple confirmations of gene importance and ultimately results in higher ranks for the driver genes. Fig. 2(c) demonstrates that the NetWalkRank algorithm consistently yields higher ranks for driver genes compared to other methods. This improvement in driver gene ranks is not limited to the final rank obtained through the weighted mean of genes in different layers. In fact, the ranks of driver genes within each individual layer of the multiplex GRN (as shown in Fig. 2(d)) are also enhanced, surpassing the ranks of non-driver genes. This finding highlights the effectiveness of NetWalkRank in prioritizing CDGs across multiple layers of the multiplex GRNs.

B. Classification results

To demonstrate the discriminatory power of the obtained ranks in CDGs, we utilized the gene ranks within each layer and the final assigned ranks obtained through various strategies as decision values for classifying driver genes and normal genes. Supplementary Section S3 provides further information on the feature selection process. In order to conduct a comparative study, we applied different strategies that leverage properties of gene expression data individually within the NetWalkRank framework. To establish a baseline, we randomly selected an equal number of normal genes and determined their ranks using different strategies. This experiment was repeated 1000 times, and performance metrics for classification were obtained. Fig 3 presents the mean receiver operating characteristic (ROC) curves and box plots displaying the distribution of the area under the curve (AUC) values.

In Fig 3, when NetWalkRank is applied to the multiplex GRNs and the LFC of each gene is used as a coefficient in each step ('DEGmlwpFC' strategy), it achieves a mean AUC value of 0.922 in classification performance. By considering the values before applying the LFC as the final value and

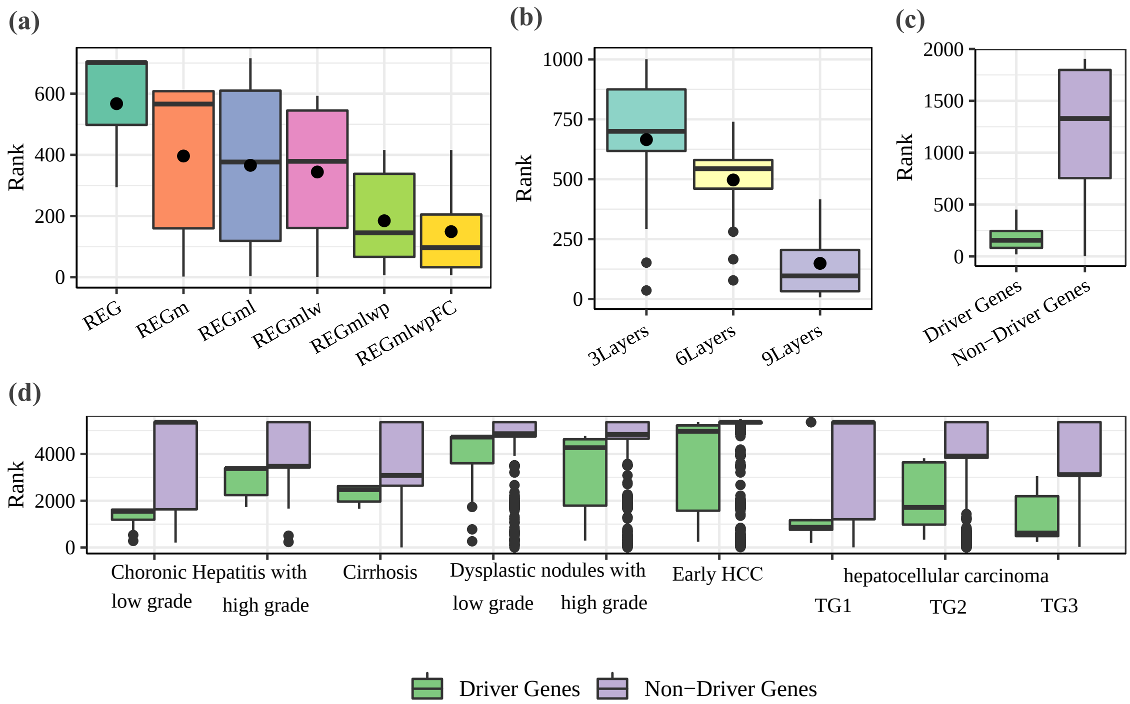

Fig. 2: The ranks for HCC driver genes by different strategies. (a) Boxplot of the ranks of cancer driver genes in different strategies. (b) Boxplot of the ranks of cancer driver genes by integrating three, six, or nine layers of gene regulatory network information. (c) Boxplot of the ranks of driver genes and non-driver genes by best strategy. (d) Boxplot of the ranks of driver genes and non-driver genes in the layers of the multiplex network by best strategy.

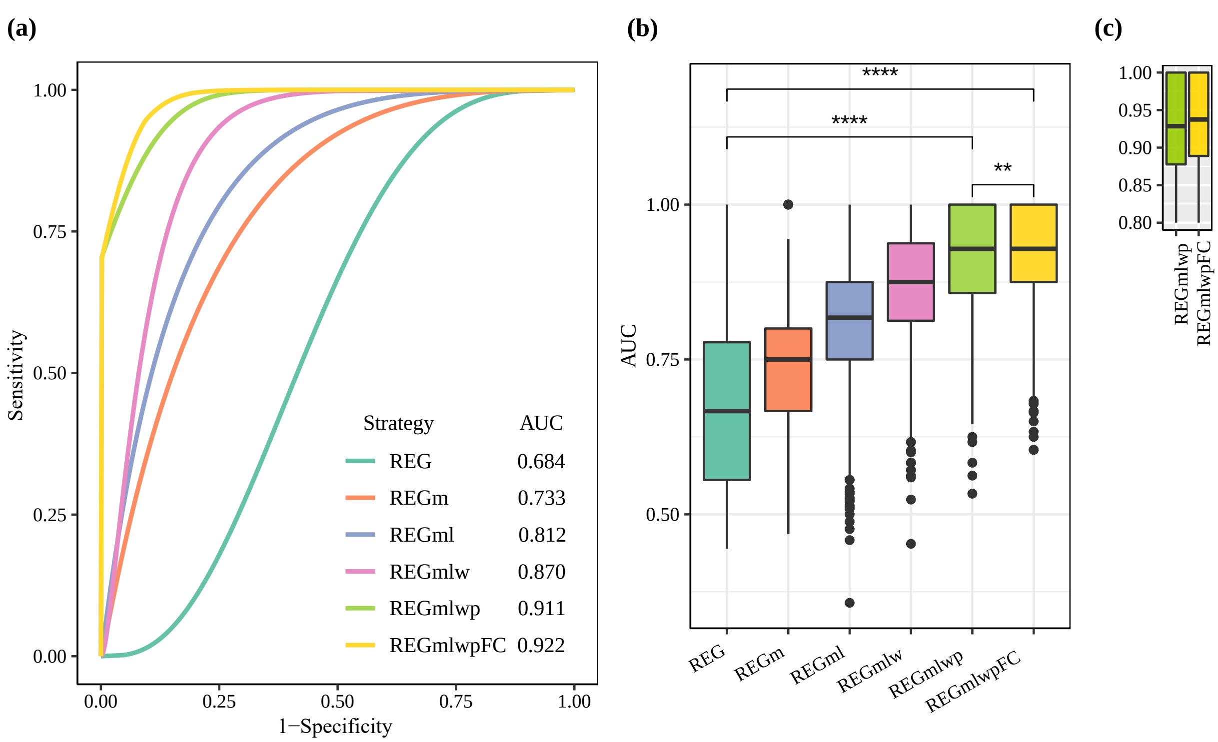

Fig. 3: The classification results of using different strategies. (a) Smoothened mean ROC curves of 1000 experiments. (b) Boxplots of AUCs in the corresponding 1000 experiments. (**** P-value $<$ 1e-4, *** P-value $<$ 1e-3, ** P-value $<$ 0.01 and * P-value $<$ 0.05 by two-side Wilcoxon test)(c) Boxplots of AUC for the last best two strategies.

TABLE I: Classification performances of different strategies, where the values are the mean ± standard deviation.

Strategies	SE	SP	ACC	F1	AUC
REG	0.920±0.142	0.658±0.210	0.789±0.103	0.814±0.090	0.685±0.161
REGm	0.958±0.076	0.507±0.169	0.730±0.101	0.777±0.085	0.733±0.092
REGml	0.781±0.165	0.843±0.156	0.811±0.102	0.797±0.117	0.812±0.102
REGmlw	0.816±0.153	0.924±0.142	0.869±0.099	0.856±0.112	0.870±0.095
REGmlwp	0.894±0.122	0.929±0.134	0.910±0.085	0.906±0.088	0.911±0.084
REGmlwpFC	**0.901±0.119**	**0.942±0.112**	**0.921±0.079**	**0.916±0.084**	**0.921±0.078**

calculating the ranks based on these values (‘DEGmlwp’), the mean AUC value for classification will be 0.911. When the initial values of genes in each layer are not set differently, the classification performance decreases to 0.870. Furthermore, in the absence of layer coefficients, the performance is 0.811. These values are notably better than the performance achieved with only the original GRN, where the mean AUC values are 0.732 for ‘REGm’ and 0.684 for ‘REG’. This demonstrates the effectiveness of integrating multistep data with multiplex networks in accurately classifying driver genes from normal genes.

Table I presents more detailed classification performance metrics for different strategies in gene prioritization from the 1000 experiments. ‘REGmlwpFC’ and ‘REGmlwp’ achieve notably high AUC values of 0.921 and 0.911, respectively, indicating their effectiveness in classification. Additionally, these strategies exhibit high sensitivity (SE) values of 0.901 and 0.893, respectively, reflecting their ability to correctly identify driver-related genes. On the other hand, ‘REGm’ demonstrates a high SE value of 0.958, but it shows lower specificity (SP) with a value of 0.50. Furthermore, it yields a relatively lower F1 score of 0.776, accuracy (ACC) of 0.729, and AUC of 0.732. These results suggest that while ‘REGm’ performs well in identifying driver genes, it may also classify some non-driver genes as driver genes. Overall, these findings provide substantial evidence for the effectiveness of integrating multistep data in prioritizing cancer genes, particularly demonstrated by ‘REGmlwpFC’ and ‘REGmlwp’ strategies.

Based on the AUC values, it is evident that the strategy of treating multistep data as a multiplex network enhances the classification outcomes. The inclusion of a coefficient based on the weight of connections in each layer within the transition matrix proves to be an effective strategy for improving classification accuracy. Notably, personalized PR demonstrates the most significant improvement in classification results. The identification and prioritization of driver genes through this approach hold promise for their potential utilization as specific diagnostic biomarkers for HCC [53].

C. Robustness evaluation in diverse network settings

Three distinct PPI networks, namely STRINGv11 [41], HumanNet [43], and InBioMap [42], were utilized to assess the robustness and generalizability of our proposed NetwalkRank across different network topologies. The purpose of employing PPI networks was not to directly compare them with GRNs but to evaluate whether our method maintains reliable performance across diverse biological networks.

The STRINGv11 network contains 11,039,526 interactions, encompassing both direct (physical) and indirect (functional) associations [41]. HumanNet consists of 977,495 interactions, comprising predicted and validated interactions [43]. InBioMap collects 598,371 interactions and is often widely employed for interpreting and visualizing biomedical big data within the contexts of systems biology [42].

To evaluate the method’s adaptability, we employed the NetWalkRank method by utilizing PPI networks in place of the RegNetwork during the initial stages of the framework. Since real-world biological networks are often incomplete and noisy, testing our method on alternative network structures helps verify its robustness in ranking disease-related genes under varying conditions. It is important to note that the networks used within NetWalkRank were incomplete. To test its robustness, we introduced perturbations by randomly introducing 30,000, 60,000 and 90,000 additional edges to the GRNs. This perturbation analysis further examined whether the method remains stable under network modifications.

Fig. 4 showcases the results obtained from utilizing different networks within the NetWalkRank method. In Fig. 4(a), the ranks of CDGs under various network mappings are presented. Notably, the driver genes consistently achieved higher ranks compared to non-driver genes, irrespective of the PPI network utilized. In most networks, the mean ranks of driver genes were relatively similar, except for InBioMap, where the mean ranks of driver genes were unexpectedly lower than those of non-driver genes. To further evaluate the performance, we utilized three criteria: precision (Pre), recall (Re, equal to SE) and F1, which combine precision and recall. Fig. 4(b) illustrates the F1, precision, and recall scores for NetWalkRank when employing STRINGv11, HumanNet, InBioMap, and RegNetwork with randomly added connections. The results demonstrate consistent patterns across these networks. HumanNet and InBioMap exhibited the lowest values for these metrics, with precision scores of 0.927 and 0.923, recall scores of 0.803 and 0.814, and F1 scores of 0.848 and 0.853, respectively. On the other hand, RegNetwork attained the highest scores, with precision, recall and F1 scores of 0.951, 0.895, and 0.914, respectively (Table II).

Furthermore, we applied different strategies using STRINGv11, HumanNet, and InBioMap as the mapping networks, which exhibited similar behavior to when RegNetwork was used in the NetWalkRank steps for constructing the multiplex network. In all cases, setting the initial seed values based on the present layer and utilizing LFC as a coefficient for the PR values before ranking the nodes resulted in improved AUC values. These findings

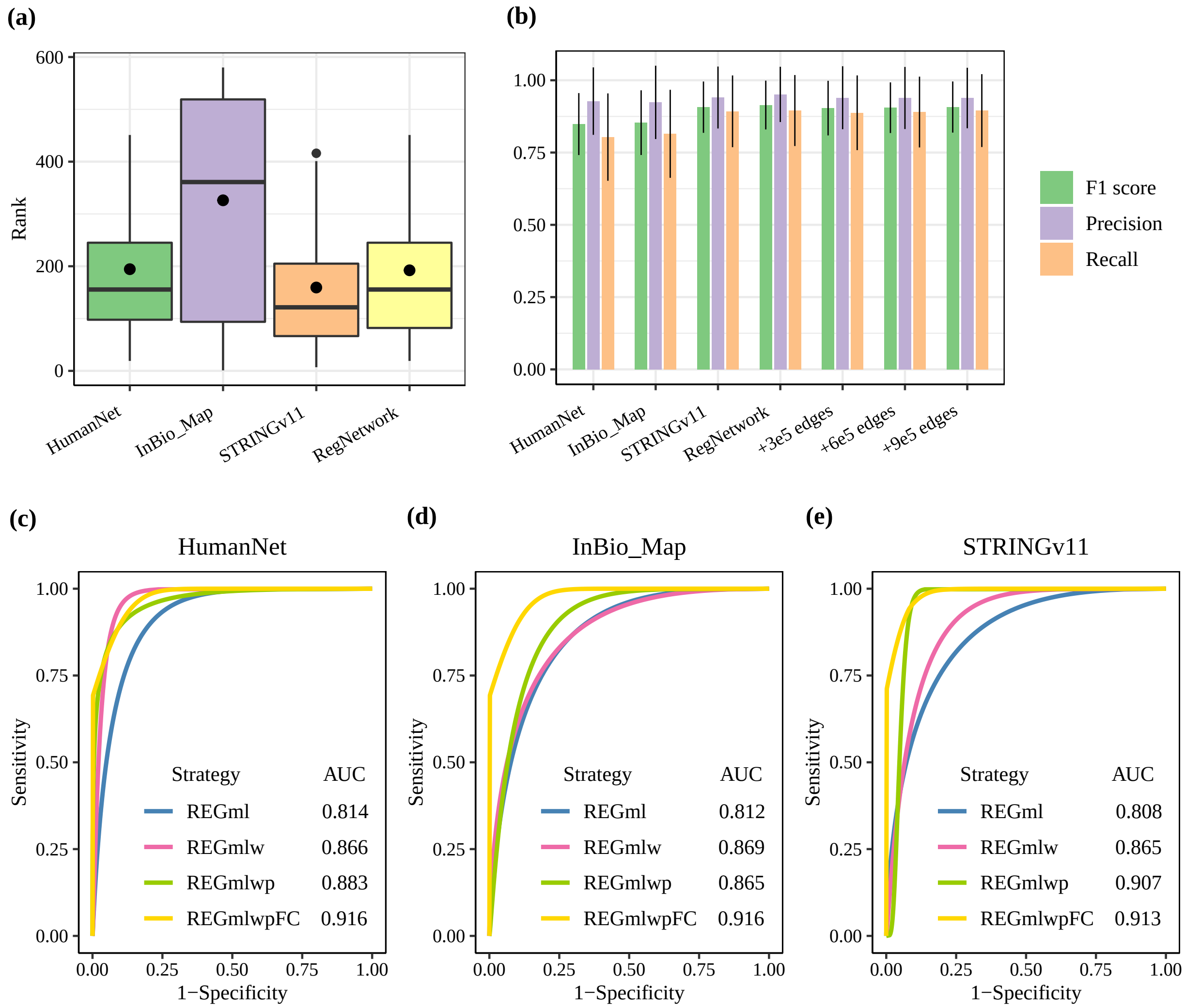

Fig. 4: Overall performance results of different network perturbations. (a) Boxplot of the ranks of CDGs by different networks mapping. (b) The median of F1, precision and recall score of four network and three network perturbations. Classification results of using different strategies in the network of (c) HumanNet, (d) InBioMap, (e) STRINGv11.

TABLE II: Classification performances on different networks.

Networks	Pre	Re	ACC	F1	AUC
HumanNet	0.928±0.116	0.803±0.151	0.864±0.093	0.849±0.107	0.864±0.093
InBioMap	0.924±0.126	0.815±0.152	0.865±0.100	0.853±0.111	0.866±0.099
STRINGv11	0.940±0.107	0.893±0.124	0.912±0.082	0.907±0.088	0.913±0.081
RegNetwork	**0.901±0.119**	**0.942±0.122**	**0.921±0.079**	**0.916±0.084**	**0.921±0.078**

confirm that while performance varies across networks, NetWalkRank consistently ranks driver genes effectively, supporting its applicability across different biological datasets.

D. Comparisons results with other methods

So far, alternative methods for prioritizing driver genes, including DawnRank [36], DriverNet [37], iMaxDriver [38] and MDPFinder [17], are available. To demonstrate the superiority of NetWalkRank, a comparative analysis was conducted against these existing methods. Fig. 5 illustrates the mean ROC curves and box plots of AUC values obtained from 1,000 experiments involving these methods, while Table III provides a comprehensive comparison of the results using different indices. Based on the classification outcomes, NetWalkRank exhibits a slight improvement over the well-known DawnRank method (AUC=0.852) and significantly outperforms the other methods. However, it should be noted that iMaxDriver achieves a high SP value of 0.93, but its SE is relatively low at 0.49, with an AUC of 0.68. On the other hand, DriverNet and MDPFinder yield the lowest AUC values among the compared methods, scoring 0.59 and 0.55, respectively.

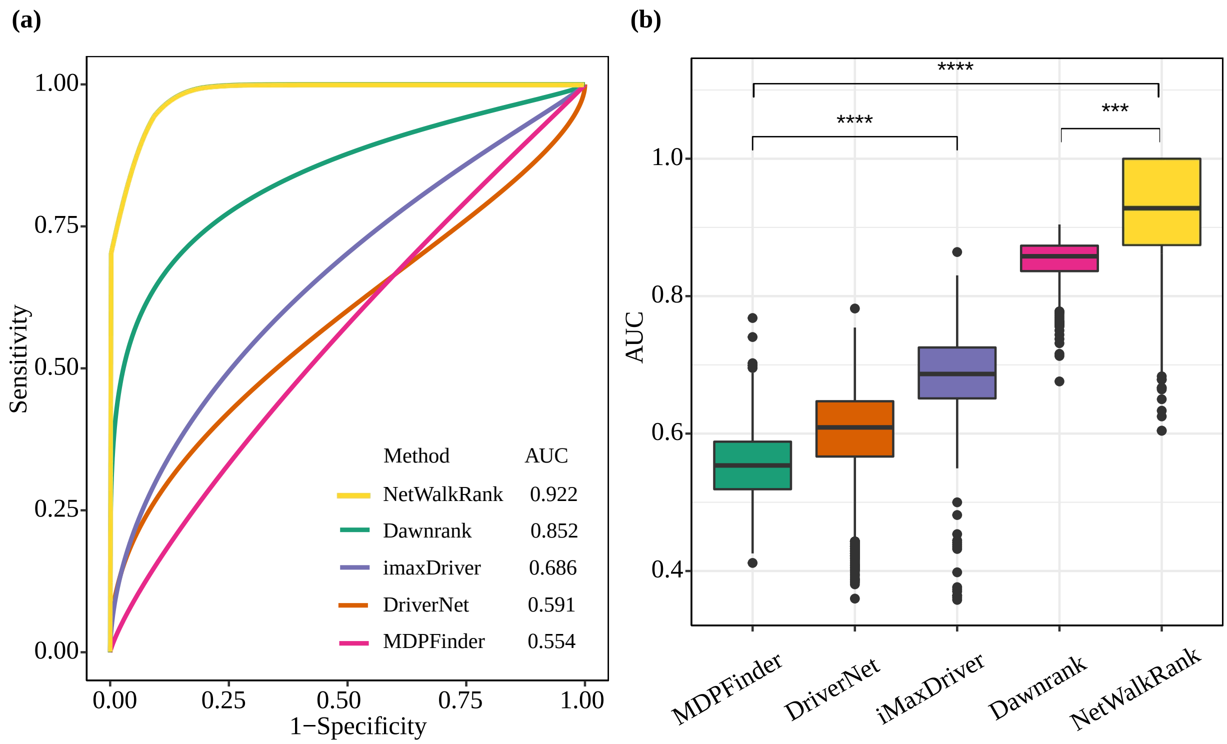

Fig. 5: ROC curves and boxplots of AUC values achieved by different methods. (a) Smoothened mean ROC curves of 1,000 experiments. (b) Boxplots of AUCs in the corresponding 1,000 experiments. (**** P-value <1e-4, *** P-value $<$ 1e-3, **P-value $<$ 0.01 and * P-value $<$ 0.05 by two-side Wilcoxon test)

TABLE III: Classification performances of different methods.

Strategies	**SE**	**SP**	**ACC**	**F1**	**AUC**
MDPFinder	0.589±0.196	0.640±0.213	0.615±0.035	0.586±0.100	0.554±0.050
DriverNet	0.420±0.129	0.885±0.165	0.653±0.047	0.539±0.070	0.591±0.083
iMaxDriver	0.496±0.117	0.931±0.122	0.714±0.026	0.628±0.048	0.686±0.063
DawnRank	0.803±0.047	0.925±0.067	0.864±0.026	0.856±0.025	0.852±0.030
NetWalkRank	**0.901±0.119**	**0.942±0.112**	**0.921±0.079**	**0.916±0.084**	**0.921±0.078**

E. Independent data validation

To verify the data independence of NetWalkRank, we conducted the entire process using a different set of multistep HCC gene expression data from the NCBI GEO database [44] under the entry ID GSE6764 [54]. The same procedures were applied to this independent second dataset. Fig. 6 displays the mean ROC curves and boxplots of AUC values, indicating the classification performance with the final obtained values. Each color in Fig. 6 represents a different strategy for prioritizing driver genes in the second HCC data.

Moreover, Table IV gives the classification metrics of NetWalkRank applied to the data. Based on the observations from Fig. 6 and Table IV, it is evident that the 'DEGmlwpFC' strategy achieves a higher AUC value compared to other strategies. These findings validate the effectiveness of NetWalkRank in prioritizing CDGs through data integration. Notably, the results emphasize the importance of analyzing multistep cancer data as a multiplex GRN.

IV. Conclusion

In this paper, we have proposed NetWalkRank to predict and prioritize CDGs from mapped gene expression data on a multiplex GRN. The utilization of multiple gene expression data offers cross-level validations for cancer variants, enhancing the accuracy and confidence in prioritizing driver genes.

In particular, we analyzed tumor samples and controls to extract DMIs and utilized them to assign weights to the relevant gene networks. Subsequently, we enhanced the PR algorithm by incorporating constraints such as genetic flow, prior knowledge, and careful consideration. We integrated the importance of each layer to determine the likelihood of genetic information transfer among genes. Furthermore, the probability of transitioning to another layer is influenced by the importance of the current and target layers. This elevates the likelihood of traversing significant edges and reaching crucial nodes, resulting in an increased PR score for those vital nodes. In our case studies, we prioritized CDGs in HCC, a typical multistep disease, using data from interactome and transcriptome. Additionally, we performed a case study with another HCC gene expression dataset to showcase the

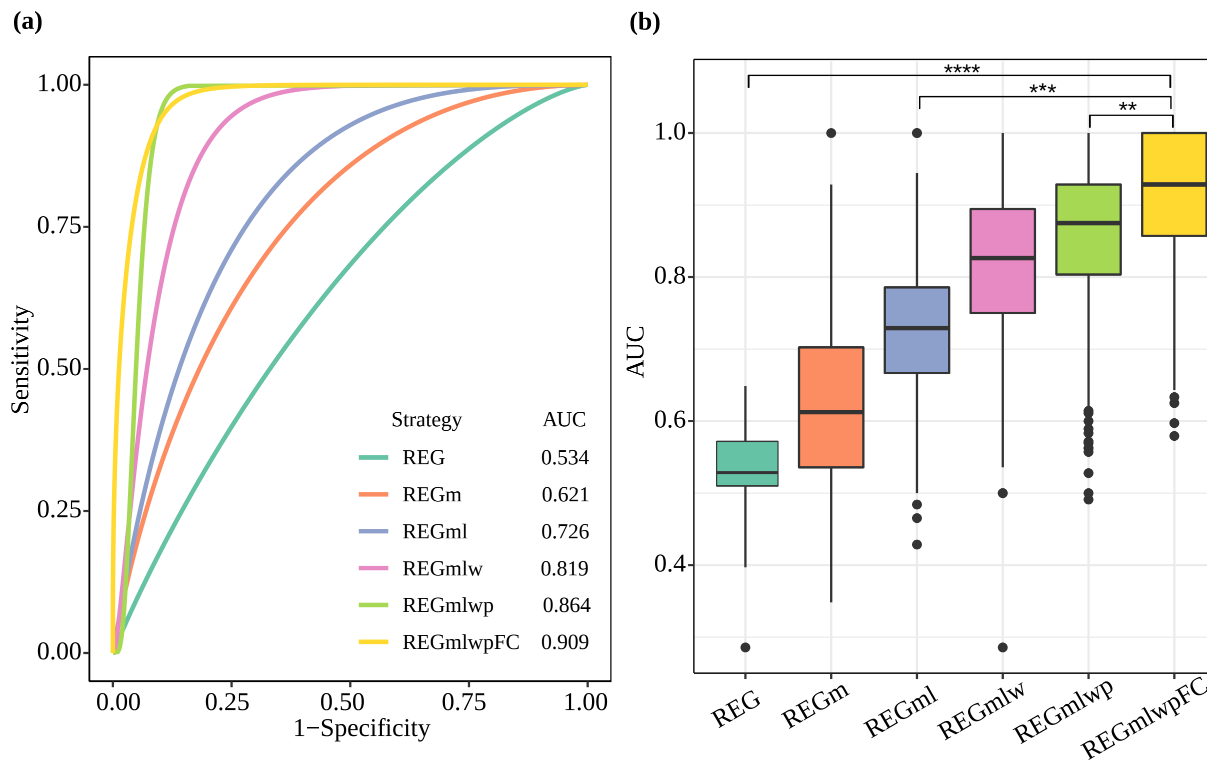

Fig. 6: ROC curves and boxplots of AUC values of different strategies in an HCC data to independency study. (a) Smoothened mean ROC curves of 1,000 experiments. (b) Boxplots of AUCs in the corresponding 1000 experiments. (**** P-value <1e-4, *** P-value < 1e-3, **P-value < 0.01 and * P-value < 0.05 by two-side Wilcoxon test)

TABLE IV: Classification performances of different strategies for independence test. Values are the mean ± standard deviation.

Strategies	SE	SP	ACC	F1	AUC
REG	0.865±0.061	0.202±0.067	0.550±0.040	0.662±0.056	0.534±0.023
REGm	0.615±0.2	0.583±0.199	0.596±0.133	0.591±0.154	0.621±0.111
REGml	0.955±0.084	0.497±0.171	0.726±0.098	0.775±0.084	0.726±0.093
REGmlw	0.779±0.164	0.859±0.150	0.818±0.101	0.803±0.118	0.819±0.101
REGmlwp	0.804±0.157	0.924±0.142	0.864±0.098	0.848±0.113	0.864±0.096
REGmlwpFC	**0.894±0.127**	**0.924±0.126**	**0.908±0.0827**	**0.903±0.089**	**0.909±0.081**

independence of our results.

To evaluate our proposed method, we included a set of genes that were commonly identified as HCC driver genes in previous studies. The rankings and classifications of these well-known driver genes confirmed the effectiveness of NetWalkRank. Additionally, comparison studies with existing methods further demonstrated the advantages of NetWalkRank. Our findings also provided evidence supporting the importance and advantage of integrating multistep gene expression data in the prioritization of CDGs on multiplex gene networks. Since different stages of the disease exhibit a diverse range of somatic mutations, it is logical to arrange them as layers within a multiplex network framework.

The inclusion of more disease stages as layers in the analysis consistently led to improved rankings for driver genes. Additionally, by incorporating additional data features in our method, the classification performance was enhanced. As anticipated, these strategies for data integration produced more informative rankings for CDGs. Notably, our evaluations demonstrated the robustness of the results across different types of networks used for gene expression mapping and the varying number of gene connections. Moreover, the versatility of the proposed NetWalkRank framework extends beyond the identification and prioritization of CDGs. The underlying network-based approach, incorporating random walking in a multiplex network, can be readily extended to enable systems biology analysis of complex diseases using multi-omics data. This flexibility facilitates the exploration of diverse biological phenomena and paves the way for comprehensive analysis across multiple domains. However, it is important to note that NetWalkRank has certain limitations. Currently, it solely relies on gene regulatory interactions, without explicitly distinguishing between upregulation and downregulation. Incorporating directionality into regulatory interactions could further enhance the biological relevance of our model. A potential future improvement involves integrating signed regulatory networks, where positive and negative weights indicate activation and repression, respectively. Additionally, leveraging curated biological knowledge could help refine the weight assignment process, making the framework more biologically interpretable. Furthermore, exploring methods to reduce the data volume required by the algorithm and refining the selection of seed nodes are potential research avenues for further investigation.

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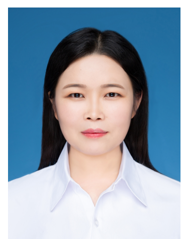

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